INCH-POUND

MIL-DTL-43649C (GL)
25 June 1997
SUPERSEDING
MIL-P-43649B (GL)
28 June 1991

# DETAIL SPECIFICATION

# PARACHUTES, CARGO, 12, 26, AND 34 FOOT DIAMETER

This specification is approved for use by the Natick Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers the requirements for three cargo parachute assemblies and the separate components thereof.

1.2 Classification.

1.2.1 Size. The cargo parachutes shall be of the following sizes as specified (see 6.2).

Size - 12-foot diameter
Size - 26-foot diameter
Size - 34-foot diameter

1.2.2 Components. The following cargo parachute components may be procured separately when specified (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Attn: SSCNC-UTE, Natick, MA 01760-5017 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1670

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For 12-foot diameter cargo parachute:
Canopy assembly
Deployment bag
Static line

For 26-foot diameter cargo parachute:
Canopy assembly
Deployment bag
Static line

For 34-foot diameter cargo parachute:
Canopy assembly
Pack
Drogue
Static line

## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL
V-T-295 - Thread, Nylon

STANDARDS

FEDERAL
FED-STD-191 - Textile Test Methods
FED-STD-751 - Stitches, Seams and Stitchings



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(Unless otherwise indicated, copies of the above specifications and standards are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094).

2.2.2 Drawings. The following Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

| | |
|---|---|
| 11-1-216 | - Parachute, Cargo, Ringslot; 12 Ft. Diameter |
| 11-1-217 | - Canopy Assembly; 12 Ft. Dia. Ringslot Cargo Parachute |
| 11-1-218 | - Deployment Bag; 12 Ft. Diameter Cargo Parachute |
| 11-1-219 | - Static Line; Cargo Parachute |
| 11-1-507 | - Canopy Details |
| 11-1-508 | - Riser; Cargo Parachute |
| 11-1-509 | - Panels, Canopy |
| 11-1-555 | - Parachute, Cargo, Ringslot; 26 Ft. Diameter |
| 11-1-556 | - Canopy Assembly; 26 Ft. Diameter Ringslot Cargo Parachute |
| 11-1-557 | - Canopy Details |
| 11-1-558 | - Canopy Sections |
| 11-1-559 | - Riser; Cargo Parachute |
| 11-1-560 | - Panels, Canopy |
| 11-1-578 | - Parachute, Cargo, 34 Ft. Nominal Diameter, Cotton, Bi-Conical Annular Ring, 500 Lb Capacity |
| 11-1-579 | - Canopy Assembly; 34 Ft. Diameter Bi-Conical, G-14 Cargo Parachute |
| 11-1-580 | - Canopy, Details and Sections |
| 11-1-581 | - Riser, 34 Ft. Diameter Bi-Conical Cargo Parachute |
| 11-1-3954 | - Deployment Bag Assembly; 26 Ft. HV Parachute |
| 11-1-3955 | - Body Assembly |
| 11-1-3956 | - Flap, Assembly |
| 11-1-3960 | - Loop Assembly |

AIR FORCE

| | |
|---|---|
| 51C6166 | - Line, Static, Parachute |
| 51D6553 | - Pack, Cargo Parachute |
| 61D2459 | - Drogue, Parachute Static Line |

(Copies of drawings are available from the U.S. Army Natick Research, Development and Engineering Center, Attn: AMSSC-S-IMI, Natick, MA 01760-5017).



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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. Materials shall be in accordance with the requirements specified on the applicable drawings (see 2.2.2), to subsidiary documents applicable thereto, and as specified herein.

3.2.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational requirements, and promotes economically advantageous life cycle costs.

3.3 Construction and assembly. The end item (see 6.2) shall conform to the requirements of this specification and to designs, details, and dimensions shown or specified on the applicable drawings.

3.3.1 Stitching. All stitching shall be in accordance with FED-STD-751. Broken threads, skipped and missing stitches occurring in type 301 or 401 stitching shall be overstitched not less than 1 inch past the ends of the defect. Broken threads and skipped stitches occurring in type 308 stitching shall be overstitched not less than three stitches past the ends of the defect. Except where otherwise specified, ends of all type 301 or 401 stitching shall be backstitched not less than 1/2 inch, unless ends are turned under in a hem or held down by other stitching. Thread tensions shall be maintained so that loose stitching does not occur and the lock will be imbedded in the materials sewed. Thread ends shall be trimmed throughout. Runoffs shall be restitched to conform to the specified stitch pattern.

3.3.1.1 Stitching, automatic machinery. Stitching patterns produced by automatic machines may be utilized in lieu of those specified, providing seam strength efficiency and function are equal to, or better than, the specified pattern, and upon approval by the procuring activity. Certification of equivalency, supported by engineering data, may be required.

3.3.1.2 Darns. Holes, cuts, or tears in the canopy, deployment bag or pack cloths shall be darned with thread, types I, II or III, class A, size A, color Camouflage Green 483 of V-T-295. Damaged areas of the cloth which can be darned shall not exceed 1/4 inch in diameter for holes (except drill holes) and 1/2 inch in length for cuts or tears in any direction. Fourteen darns are allowed per canopy, four per deployment bag, and four per pack.



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3.3.2 Canopy permeability. The air permeability of the cloth of the completed canopy shall not exceed the specified lower limit by more than 10 percent nor the specified upper limit by more than 10 percent when tested in accordance with 4.3.5.

3.3.3 Canopy patching. One sewn patch is allowed per canopy. The maximum damaged area that can be patched is an area 7 inches by 7 inches. The patch shall be located no closer than 1 inch from a radial seam, edge reinforcement, or lower lateral band. The patch shall be square or rectangular in shape and conform to the patching procedure in 3.3.4.

3.3.4 Patching procedure.

a. Place the repairable item on a repair table, smooth the fabric around the damaged area, and secure the item to the table with pushpins. Do not pin the damaged area.

b. Using an authorized marking aid of contrasting color, mark a square or rectangle around the area to be patched and insure that one side of the marked square or rectangle is parallel to the warp or filling of the fabric.

c. Cut the damaged area fabric along the lines made in (b) above. Further cut the fabric diagonally at each corner to allow a 1/2-inch foldback in the raw edges.

d. Make a 1/2-inch foldback on each raw edge. Pin and baste each foldback to complete the prepared hole.

e. Using the same type material as in the original construction, mark and cut a patch 2-1/2 inches wider and longer than the inside measurements of the prepared hole. Insure that the patch material is marked and cut along the warp or filling of the fabric.

f. Center the patch material over the prepared hole and insure the warp or filling of the patch material matches the warp or filling of the fabric being patched. Pin the patched material in position.

g. Make a 1/2-inch fold under on each edge of the patch material and baste the patch to the prepared area.

h. Remove the pushpins securing the item to the repair table and secure the patch by machine stitching, using the applicable details in figure 1. The stitching should be accomplished with thread that is contrasting in color to the fabric. Thread shall be V-T-295, size E nylon, types



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I, II or III, class A. The stitching shall be type 301, using 7 to 11 stitches per inch in accordance with FED-STD-751. Backstitch ends of stitching 1/2 inch minimum or overlap the ends of stitching 2 inches minimum or 4 inches maximum.

3.4 Workmanship. The workmanship shall conform to the quality of product established by this specification.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2)
b. Conformance inspection (see 4.3)

4.2 First article inspection. When a first article inspection is required (see 3.1 and 6.2) it shall be examined in accordance with 4.3.

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.3.1, 4.3.2, 4.3.3 and 4.3.4 and the testing of 4.3.5. Sampling for the inspections in 4.3.3, 4.3.4, and 4.3.5 shall be as specified in the contract or purchase order (see 6.2).

4.3.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.3.2 In-process inspection. Inspection of subassemblies shall be made to ascertain that construction details and dimensional requirements which cannot be examined in the finished product are in accordance with specified requirements. The Government reserves the right to exclude from consideration for acceptance, any material or service for which in-process inspection has indicated nonconformance.

4.3.3 End item visual examination.



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4.3.3.1 Canopy. The canopies shall be examined for the applicable defects listed in table I. Each noncompliance shall be scored as a defect.

TABLE I. Canopy visual defects

| Defect | Classification | |
|---|---|---|
| | Major | Minor |
| Cloth, webbing and tape: | | |
| Any hole in cloth (except drill holes): | | |
| 1/4 inch diameter or less | | 201 |
| More than 1/4 inch diameter | 101 | |
| Color not as specified | | 202 |
| Not specified type or class | 102 | |
| Cut or tear in cloth: | | |
| 1/2 inch or less | | 203 |
| More than 1/2 inch | 103 | |
| Smash | 104 | |
| Multiple floats | 105 | |
| Grease or oil stain over 3/4 sq. inch | 106 | |
| Grease or oil stain(s) less than 3/4 sq. inch area: | | |
| Not more than 5 per gore or 15 per canopy | | 204 |
| More than 5 per gore or 15 per canopy | 107 | |
| Embedded crease or wrinkle in excess of 1/2 inch width | 108 | |
| Darns, over 14 per canopy | 109 | |
| Patches: | | |
| More than one per canopy | 110 | |
| Patch size exceeding 7 inches by 7 inches | 111 | |
| Location not as specified | 112 | |
| Shape not as specified | 113 | |
| Procedure, not as specified in 3.3.4 | 114 | |
| Cord and thread: | | |
| Not specified type or size as applicable | 115 | |
| Color not as specified | | 205 |



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TABLE I. Canopy visual defects (contd)

| Defect | Classification Minor | Major |
|---|---|---|
| Minor | | |
| Cloth, webbing and tape: | | |
| Vent reinforcement overlap, does not span two main seams | 116 | |
| Vent reinforcement band, improper assembly | 117 | |
| Vent line length (across vent) not as specified: | | |
| Less than 1/4 inch | 118 | |
| 1/4 inch or more | | 206 |
| Twisted vent line: | | |
| More than $360^0$ | | 207 |
| More than $360^0$, causing kinks in line | 119 | |
| Gore width at vent, measured C/L to C/L on adjacent radial seams, not as specified: | | |
| Less than 1/4 inch | | 208 |
| 1/4 inch or more | 120 | |
| Slot width, not as specified: | | |
| Less than 1/8 inch | | 209 |
| 1/8 inch or more | 121 | |
| Skirt reinforcement band improperly assembled | 122 | |
| Under fold (raw edge) skirt hem | 123 | |
| NOTE: When a raw edge is the result of overfolding the edge extending more than 1/2 inch but less than 1 inch, the raw edge may be trimmed and will not be scored as a defect. If the condition extends more than 1 inch from the inside edge of the skirt hem, the stitching will be removed and the hem refolded and restitched. | | |
| Tucks, wherein section of gore fabric is caught in stitching of hems | 124 | |



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TABLE I. Canopy visual defects(contd)

| Defect | Classification | |
|---|---|---|
| | Major | Minor |
| Suspension lines crossed | 125 | |
| Cut, frayed, or damaged suspension or vent lines | 126 | |
| Cloth, webbing and tape: | | |
| Suspension line length, not as specified: | | |
| Less than 3/4 inch | | 210 |
| 3/4 inch or more | 127 | |
| Suspension line twisted: | | |
| More than $360^0$ | | 211 |
| More than $360^0$, causing kinks | 128 | |
| Stitches and stitching: | | |
| Wrong type | 129 | |
| Two or more stitches per inch less than specified | 130 | |
| Two or more stitches per inch more than specified | | 212 |
| Broken, skipped or missing stitches not overstitched as specified | 131 | |
| Open seam | 132 | |
| NOTE: A seam shall be classified as open when one or more stitches joining a seam are broken or where two or more skipped stitches occur. Repairs of open seams will not be scored as defects. | | |
| Wrong type seam | 133 | |
| Missing or incomplete stitch pattern | 134 | |
| Runoff not restitched in accordance with specified pattern | | 213 |
| Zigzag stitching not sewn through suspension lines | 135 | |
| Tension loose, resulting in loose top or bobbin thread; or tight tension, resulting in tightness or puckering of material | | 214 |
| Thread ends not trimmed | | 215 |
| Ends of stitching not secured as specified | | 216 |



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TABLE I. Canopy visual defects(contd)

| Defect | Classification | |
|---|---|---|
| | Major | Minor |
| Stitch pattern lengths: | | |
| Less than 1/8 inch minus | 136 | |
| 1/8 inch or more minus | | 217 |
| 1/2 inch or more plus | | 218 |
| Stitch pattern not complete | 137 | |
| Needle or feed dog damage | 138 | |
| Length of components and location dimensions not otherwise covered, not as specified: | | |
| Up to 5 inches: | | |
| Less than 1/8 inch | | 219 |
| 1/8 inch or more | 139 | |
| 5 inches or over: | | |
| Less than 1/4 inch | | 220 |
| 1/4 inch or more | 140 | |
| Riser: | | |
| Length overall, not as specified by: | | |
| Less than 1/2 inch | | 221 |
| 1/2 inch or more | 141 | |
| Loop length, not as specified by: | | |
| Less than 1/8 inch | | 222 |
| 1/8 inch or more | 142 | |
| Insertion of suspension line in core at skirt attachment loop and riser attachment point not as specified by: | | |
| Less than 1/4 inch | | 223 |
| 1/4 inch to 1/2 inch | 143 | |
| More than 1/2 inch plus | | 224 |
| Ends of webbing and tape not seared as specified | | 225 |
| Suspension line attachment loop length at skirt band not as specified by: | | |
| Less than 1/8 inch | | 226 |
| 1/8 inch or more | 144 | |



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TABLE I. Canopy visual defects(contd)

| Defect | Classification | |
|---|---|---|
| | Major | Minor |
| Any component missing | 145 | |
| Any incomplete assembly | 146 | |
| Identification marking illegible, incomplete, or missing | | 227 |
| Clevis: | | |
| Component missing | 147 | |
| Part fractured, malformed, or damaged | 148 | |
| Corrosion or rust | 149 | |

4.3.3.2 Deployment bag. The deployment bags shall be examined for the applicable defects listed in table II. Each noncompliance shall be scored as a defect.

TABLE II. Deployment bag visual defects

| Defect | Classification | |
|---|---|---|
| | Major | Minor |
| Cloth, webbing and tape: | | |
| Any hole in cloth (except drill holes): | | |
| 1/4 inch diameter or less | | 201 |
| More than 1/4 inch diameter | 101 | |
| Color not as specified | | 202 |
| Not specified type or class | 102 | |
| Cut or tear in cloth: | | |
| 1/2 inch or less | | 203 |
| More than 1/2 inch | 103 | |
| Darns, over 4 per bag | 104 | |
| Thread: | | |
| Not specified type or size | 105 | |
| Color not as specified | | 204 |



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TABLE II. Deployment bag visual defects (contd)

| Defect | Classification Major | Minor |
|---|---|---|
| Finished dimensions, body and flap (prior to assembly): | | |
| Less than 1/4 inch less than specified | | 205 |
| 1/4 inch or more less than specified | 106 | |
| More than 1/2 inch more than specified | | 206 |
| Stitches and stitchings: | | |
| Wrong type | 107 | |
| Two or more stitches per inch less than specified | 108 | |
| Two or more stitches per inch more than specified | | 207 |
| Broken, skipped, or missing stitches not overstitched as specified | 109 | |
| Open seam | 110 | |
| NOTE: A seam shall be classified as open when one or more stitches joining a seam are broken or where two or more skipped stitches occur. Repairs of open seams will not be scored as defects. | | |
| Runoff not restitched in accordance with specified pattern | | 208 |
| Tension loose, resulting in exposed top or bobbin thread, or tight tension, resulting in tightness or puckering of material | | 209 |
| Ends of stitching not secured as specified | | 210 |
| Stitch pattern lengths not as specified by: | | |
| Less than 1/8 inch | | 211 |
| 1/8 inch or more | 111 | |
| Thread ends not trimmed | | 212 |
| Stitch pattern not complete | 112 | |
| Needle or feed dog damage | 113 | |
| Length of components and location dimensions, not otherwise covered, not as specified: | | |
| Up to 5 inches: | | |
| Less than 1/8 inch | | 213 |
| 1/8 inch or more | 114 | |



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TABLE II. Deployment bag visual defects (contd)

| Defect | Classification | |
|---|---|---|
| | Major | Minor |
| Less than 1/4 inch | | 214 |
| 1/4 inch or more | 115 | |
| Partial assembly or any component missing | 116 | |
| Misassembly | 117 | |
| Marking illegible, incorrect or missing | | 215 |

4.3.3.3 Pack. The packs shall be examined for the applicable defects listed in table III. Each noncompliance shall be scored as a defect.

TABLE III. Pack visual defects

| Defect | Classification | |
|---|---|---|
| | Major | Minor |
| Cloth, webbing and tape: | | |
| Any hole in cloth (except drill holes): | | |
| 1/4 inch diameter or less | | 201 |
| More than 1/4 inch diameter | 101 | |
| Color not as specified | | 202 |
| Not specified type or class | 102 | |
| Cut or tear in cloth: | | |
| 1/2 inch or less | | 203 |
| More than 1/2 inch | 103 | |
| Darns, over 4 per pack | 104 | |
| Thread: | | |
| Not specified type or size | 105 | |
| Color not as specified | | 204 |



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TABLE III. Pack visual defects (Contd)

| Defect | Classification Major | Minor |
|---|---|---|
| Finished dimensions of panel assemblies (prior to assembly): | | |
| Less than 1/4 inch less than specified | | 205 |
| 1/4 inch or more less than specified | 106 | |
| More than 1/2 inch more than specified | | 206 |
| Stitches and stitchings: | | |
| Wrong type | 107 | |
| Two or more stitches per inch less than specified | | 207 |
| Two or more stitches per inch more than specified | 108 | |
| Broken, skipped, or missing stitches not overstitched as specified | 109 | |
| Open seam | 110 | |
| NOTE: A seam shall be classified as open when one or more stitches joining a seam are broken or where two or more skipped stitches occur. Repairs of open seams will not be scored as defects. | | |
| Runoff not restitched in accordance with specified pattern | | 208 |
| Tension loose, resulting in exposed top or bobbin thread or tight tension, resulting in tightness or puckering of material | | 209 |
| Ends of stitching not secured as specified | | 210 |
| Stitch pattern lengths not as specified by: | | |
| Less than 1/8 inch | | 211 |
| 1/8 inch or more | 111 | |
| Thread ends not trimmed | | 212 |
| Missing or incomplete stitch pattern | 112 | |
| Needle or feed dog damage | 113 | |
| Length of components and location dimensions, not otherwise covered, not as specified: | | |



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TABLE III. Pack visual defects (contd)

| Defect | Classification Major | Minor |
|---|---|---|
| Up to 5 inches: | | |
| Less than 1/8 inch | | 213 |
| 1/8 inch or more | 114 | |
| 5 inches or more: | | |
| Less than 1/4 inch | | 214 |
| 1/4 inch or more | 115 | |
| Partial assembly or any component missing | 116 | |
| Misassembly | 117 | |
| Marking illegible, incorrect, or missing | | 215 |

4.3.3.4 Static line . The static line shall be examined for the applicable defects listed in table IV. Each noncompliance shall be scored as a defect.

TABLE IV. Static line visual defects

| Defect | Classification Major | Minor |
|---|---|---|
| Webbing: | | |
| Any hole, cut, tear or abrasion | 101 | |
| Not specified type or class | | 201 |
| Color not as specified | 102 | |
| Length of static line not as specified by: | | |
| Up to 1/2 inch | | 202 |
| 1/2 inch or more | 103 | |
| Loop length: | | |
| Up to 1/4 inch less than specified | | 203 |
| 1/4 inch or more less than specified | 104 | |
| More than 1/4 inch more than specified | | 204 |
| Component missing | 105 | |



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TABLE IV. Static line visual defects (contd)

| Defect | Classification Major | Minor |
|---|---|---|
| Stitching: | | |
| Wrong type | 106 | |
| Missing or incomplete stitch pattern | 107 | |
| Runoff not restitched in accordance with specified stitch pattern | | 205 |
| Tension loose, resulting in loose top or bobbin thread, or tight tension, resulting in tightness or puckering of material | | 206 |
| Ends of stitching not secured as specified | | 207 |
| Broken, skipped, or missing stitches not overstitched as specified | 108 | |
| Open seam | 109 | |
| NOTE: A seam shall be classified as open when one or more stitches joining a seam are broken or where two or more skipped stitches occur. Repairs of open seams will not be scored as defects. | | |
| Stitch pattern lengths not as specified by: | | |
| Less than 1/8 inch | | 208 |
| 1/8 to 1/4 inch | 110 | |
| 1/4 inch or more plus | | 209 |
| Thread ends not trimmed | | 210 |
| Ends of webbing not seared | | 211 |
| Thread: | | |
| Not specified type or size | 111 | |
| Color not as specified | | 212 |
| Identification marking illegible, missing or incorrect | 213 | |
| Clevis: | | |
| Component missing | 112 | |
| Any part fractured, malformed, or damaged | 113 | |

4.3.3.5 Drogue. The drogues shall be examined for the applicable defects listed in Table V. Each noncompliance shall be scored as a defect.



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TABLE V. Drogue visual defects

| Examine | Defect | Classification Major | Minor |
|---|---|---|---|
| Material | Wrong color | | 201 |
| | Not specified type or class | | 202 |
| Design | Any characteristic not in accordance with specified requirement | 101 | |
| Construction and assembly | Any component missing, or not assembled or joined as specified | 102 | |
| Cleanness | Grease, dirt, oil, or other foreign matter | | 203 |
| Seams | Open seam | 103 | |
| | Open seam overstitched less than 1 inch | | 204 |
| | NOTE: A seam shall be classified as open whenever breaks, runoffs, or two or more consecutive skips occur in stitching and are not over-stitched or restitched as specified. | | |
| Stitches and stitching | Not type of stitch specified | 104 | |
| | Ends of stitching not secured as specified (except where turned under in a hem or held down by other stitching) | | 205 |
| Stitches and stitching (contd) | One stitch per inch less than minimum specified | | 206 |
| | Two or more stitches per inch less than minimum specified | 105 | |
| | One or more stitches per inch in excess of maximum specified | | 207 |
| | Loose tension, resulting in loose top or bobbin thread; or tight tension resulting in tightness or puckering of material at stitch line | | 208 |
| | Thread ends not trimmed | | 209 |
| | Any row of stitching missing | 106 | |
| | Needle chews | 107 | |
| | Stitching pattern not complete | 108 | |
| | Stitch margin or gage not as specified | | 210 |
| | Stitching run off at edge cutting material | 109 | |



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TABLE V. Drogue visual defects (contd)

| Defect | | Classification Major | Minor |
|---|---|---|---|
| Nylon cloth and tape | Holes, tears, or cuts | 110 | |
| | Abrasion or thin material | 111 | |
| Nylon cord | Cut, tear, or break | 112 | |
| | Abrasion or seared area (except on ends of cord) | 113 | |
| | Any splice or not continuous | | 211 |
| | Ends not seared | | 212 |
| Identification marking | Missing, illegible, incorrect, wrong color or location, or not applied as specified | | 213 |

4.3.4 End item dimensional examination. The end items (complete parachute assemblies or separate components, as applicable) shall be examined for conformance to the dimensions specified on the drawings. Only those dimensions that can be evaluated without damaging or disassembling the end items shall be examined. Each noncompliance shall be scored as a defect.

4.3.5 End item testing. The completely fabricated canopy shall be tested as specified in 4.4.1 for conformance to the air permeability requirement of 3.3.2.

4.4 Methods of inspection.

4.4.1 Canopy permeability test. The canopy permeability shall be determined in accordance with Method 5450 of FED-STD-191, except that three permeability readings shall be taken at widely separated points in each of the alternate gores around the canopy, and the average of the results determined. Failure of the canopy to meet the requirement in 3.3.2 shall constitute failure of the test.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.



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6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory).

6.1 Intended use. The 12-foot and 26-foot diameter parachutes are intended for use in air-dropping non-fragile cargo at high rates of descent. The 34-foot diameter parachute is intended for use in air-dropping cargo at normal rates of descent.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification
b. Size parachute required (see 1.2.1)
c. Parachute component(s) required, when procured separately (see 1.2.2)
d. When a first article is required (see 3.1 and 4.2)
e. Sampling plan, lot size and sample size (see 4.2 and 4.3)
f. Packaging requirements (see 5.1)

6.3 Subject term (key word) listing.

| | |
|---|---|
| Airdrop | Deployment bag |
| Canopy | Pack |
| Drogue | Static line |

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Army - GL

Preparing activity:
Army - GL

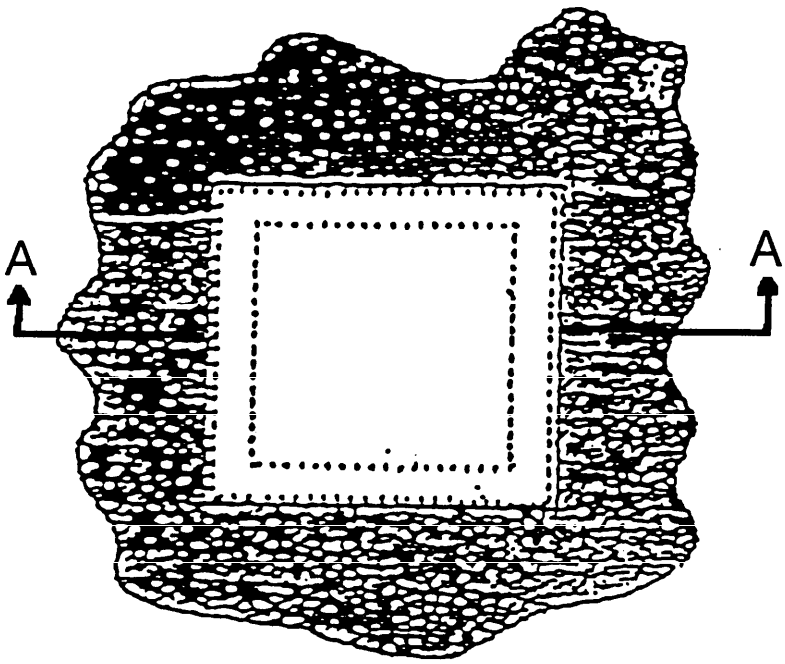
Review activity
Army - AV

(Project 1670-A887)

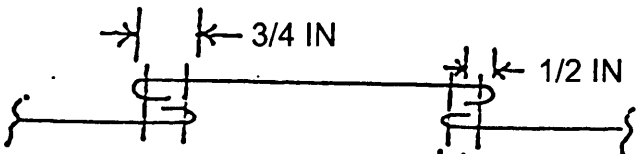
FIGURE 1. Sewn patch application to solid parachute canopy - cargo



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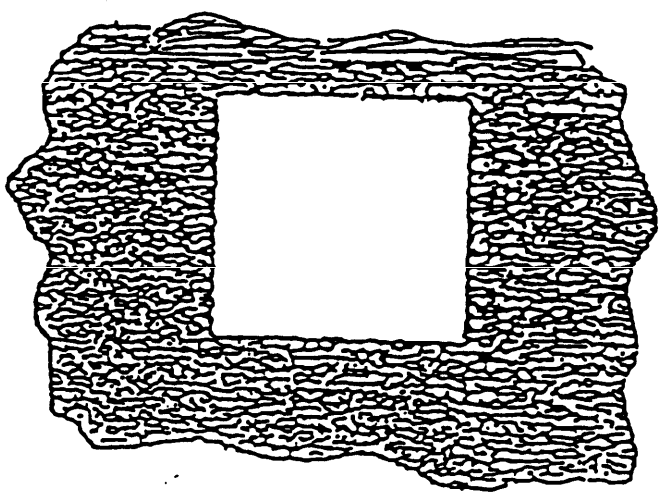

INSIDE VIEW




SECTION A-A

ALL STITCHING SHALL BE 1/8 INCH FROM EDGE

TOLORENCE: ± 1/16 IN



OUTSIDE VIEW



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

**INSTRUCTIONS**

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| | 1. DOCUMENT NUMBER | 2. DOCUMENT DATE (YYMMDD) |
|---|---|---|
| [redacted] | MIL-DTL-43649C (GL) | 97-06-25 |

3. DOCUMENT TITLE Parachutes, Cargo, 12, 26, and 34 Foot Diameter

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

[redacted]

| a. NAME | b. TELEPHONE (Include Area Code) |
|---|---|
| U.S. Army Natick RD&E Center | (1) Commercial 508-233-5175 (2) AUTOVON 256-5175 |
| c. ADDRESS (Include Zip Code)<br>Commander, U.S. Army Natick RD&E Center<br>Attn: SSCNC-WEF<br>Natick, MA 01760-5018 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 |



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